Cancer Epigenetics

Oral Midterm Exam I

Module I: Fundamentals of Epigenetics

DEC 14 2025

PHUC-LOI LUU, PHD

EMAIL: LUU.P.LOI@GOOGLEMAIL.COM

ZALO: 0901802182

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Background Concepts

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6	a	Gene Expression	mRNA/Protein	1
15	b	Genetic information	Profiling variants/mutations	2
1	c	A cellular trait	1.5% of human genome	3
7	d	RNA-seq	Histone Octamer	4
2	e	DNA-seq	loosely packed and active	5
3	f	Protein coding genes	converts genetic information into a cellular trait	6
11	g	RNA Pol I	Profiling gene expression	7
16	h	**RNA Pol II**	Always heterochromatin (e.g., centromeres)	8
4	k	**Nucleosome**	They act as regulatory neighborhoods and preventing regulatory crosstalk.	9
12	l	chromatosome	LADs	10
5	n	**Euchromatin**	Transcribes the large ribosomal RNA genes	11
14	m	**Heterochromatin**	Histone Octamer + linker H1	12
8	p	**Constitutive**	Can be reversibly converted to euchromatin (e.g., developmental genes)	13
13	q	**Facultative**	tightly packed and silenced	14
9	s	TADs	DNA	15
10	x	Tethering to the periphery is strongly associated with stable, long-term gene silencing.	Transcribes all **protein-coding** genes (mRNA)	16

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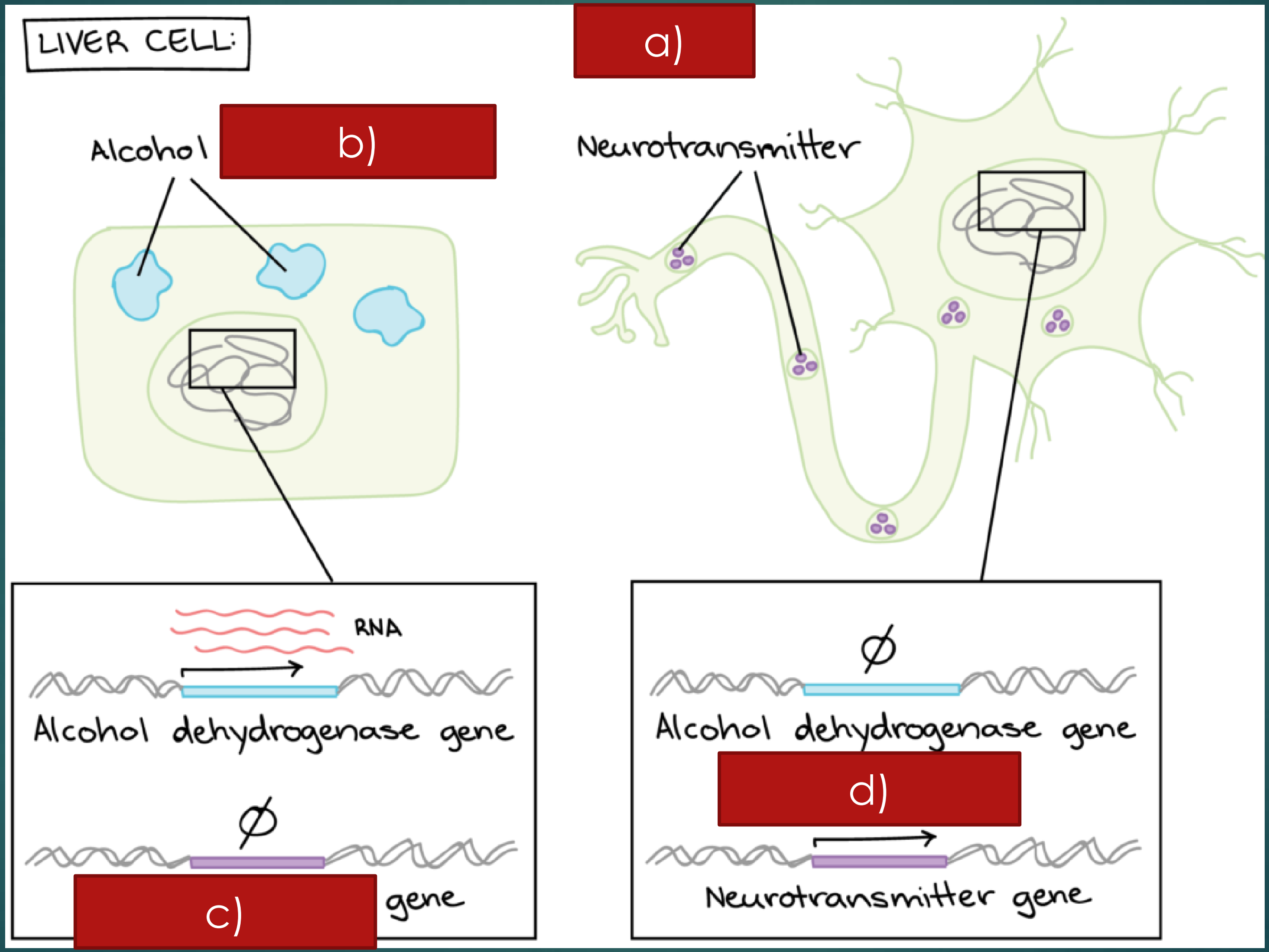
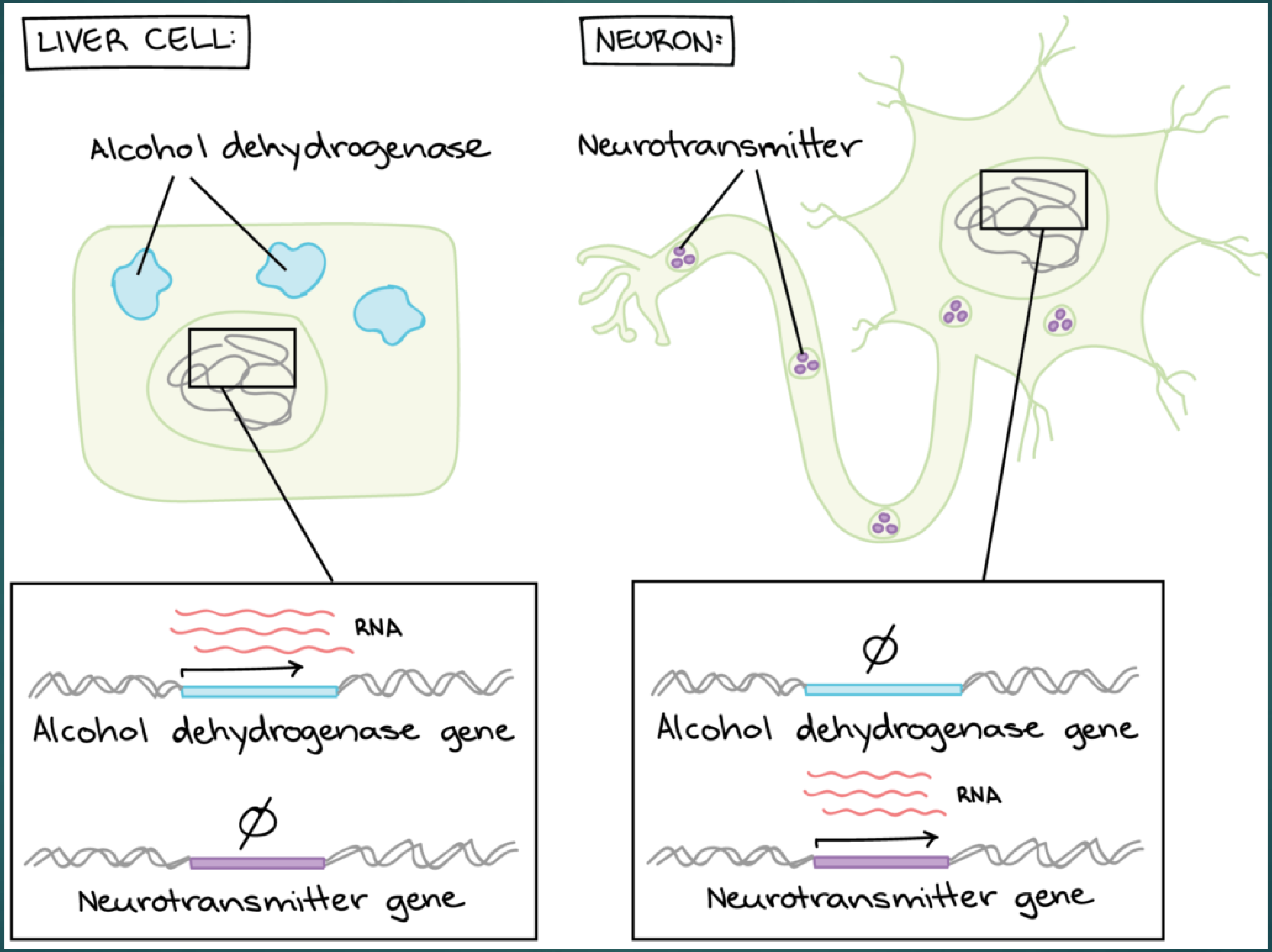
The Outcome of Gene Expression



https://courses.lumenlearning.com/wm-nmbiology1/chapter/reading-expression-of-genes/

The Outcome of Gene Expression



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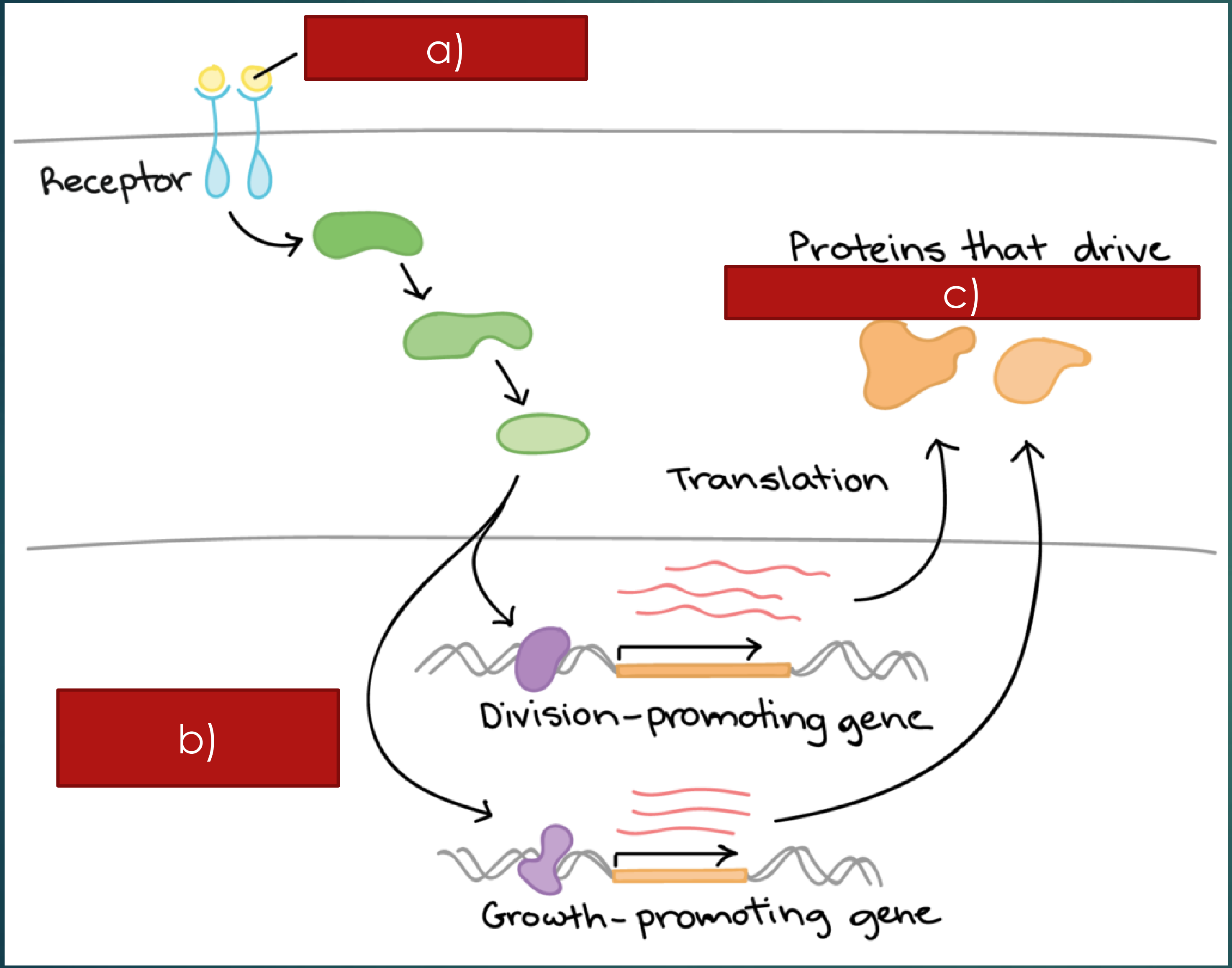
Gene expression is triggered by a signal from the cell environment



- Process of Signal Transduction Cascades
  1. An **Extracellular Signal (ligand)** binds to a cell-surface or intracellular Receptor.
  2. This **activates** a downstream cascade, often involving sequential activation of Protein Kinases.
  3. The final step **modifies a nuclear protein**: a Transcription Factor (TF) or a Chromatin Modifying Enzyme.

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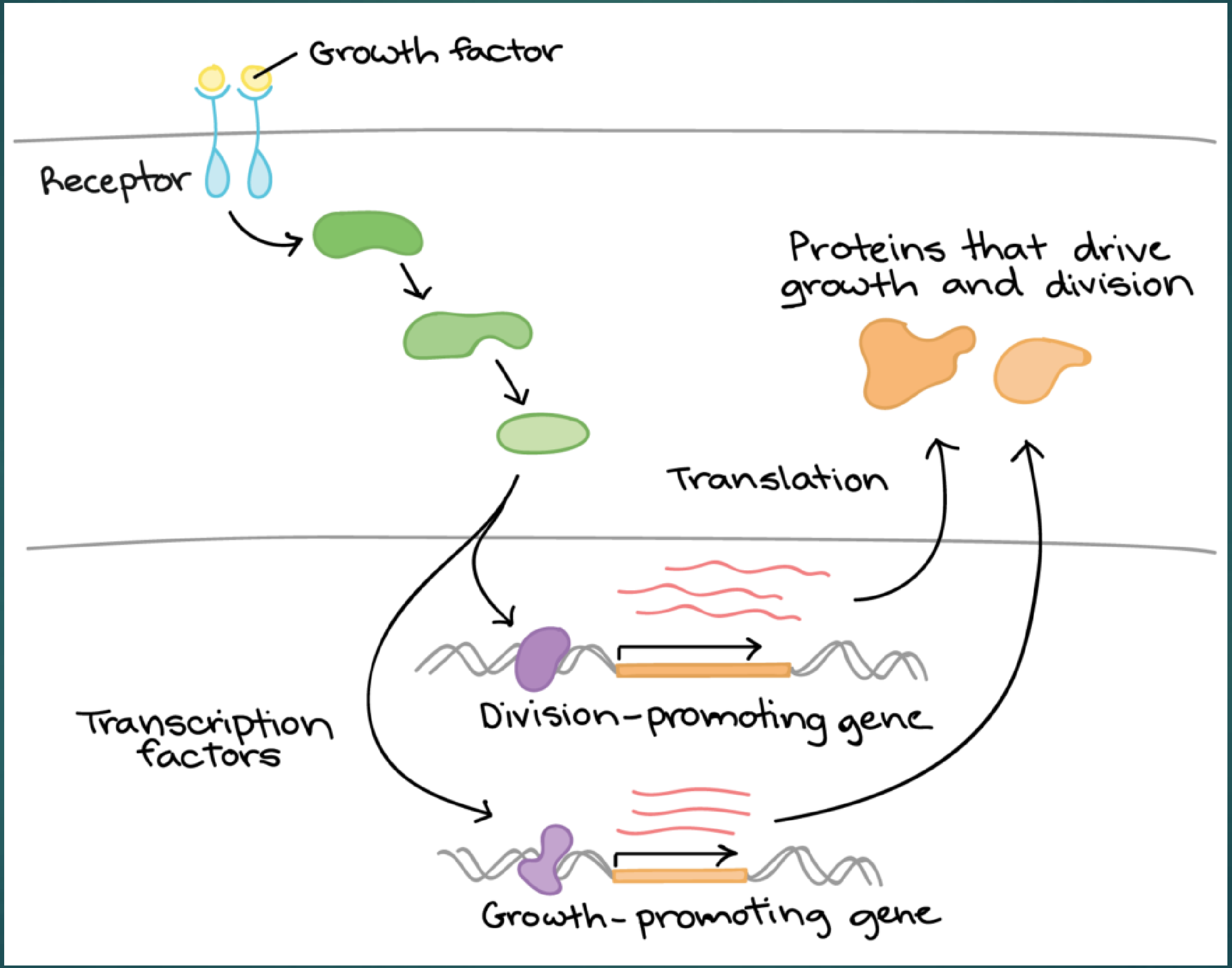

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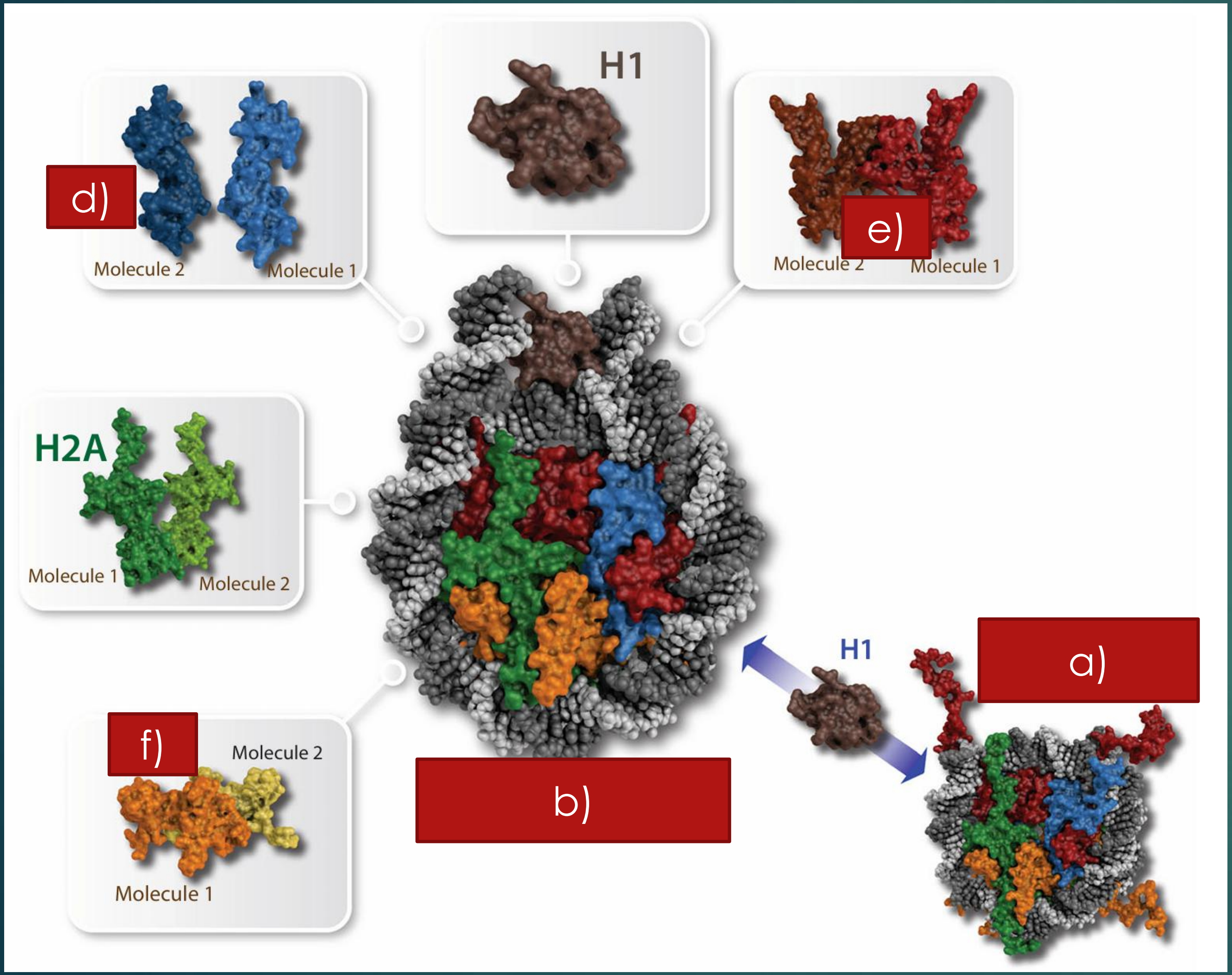
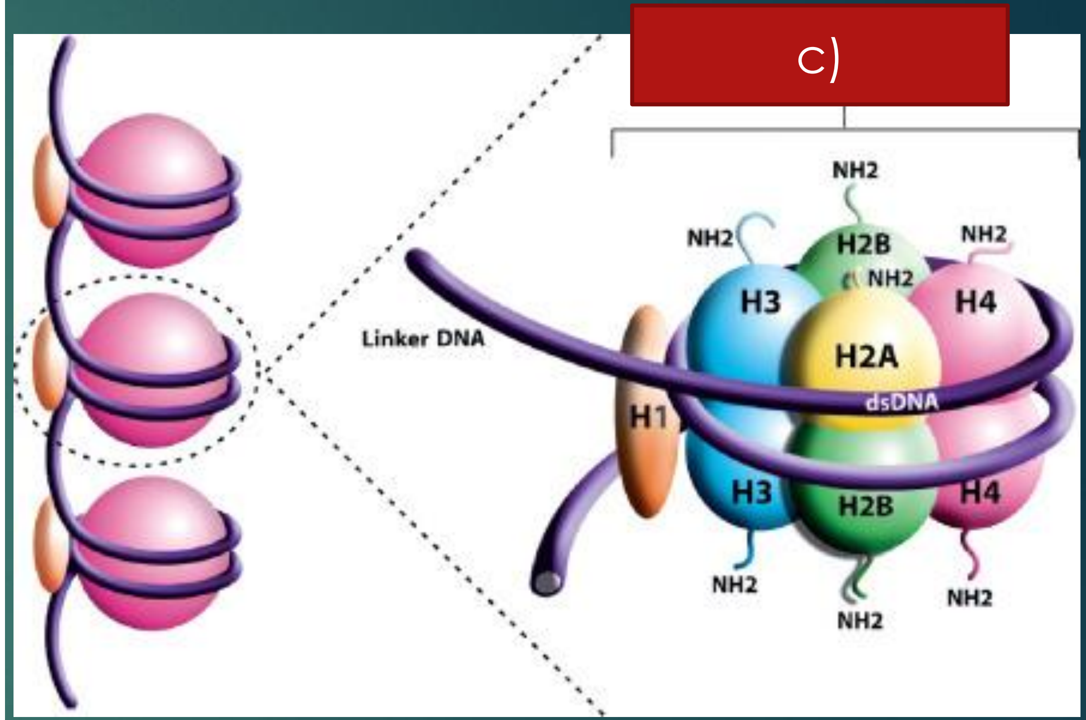
The Histone Octamer Core





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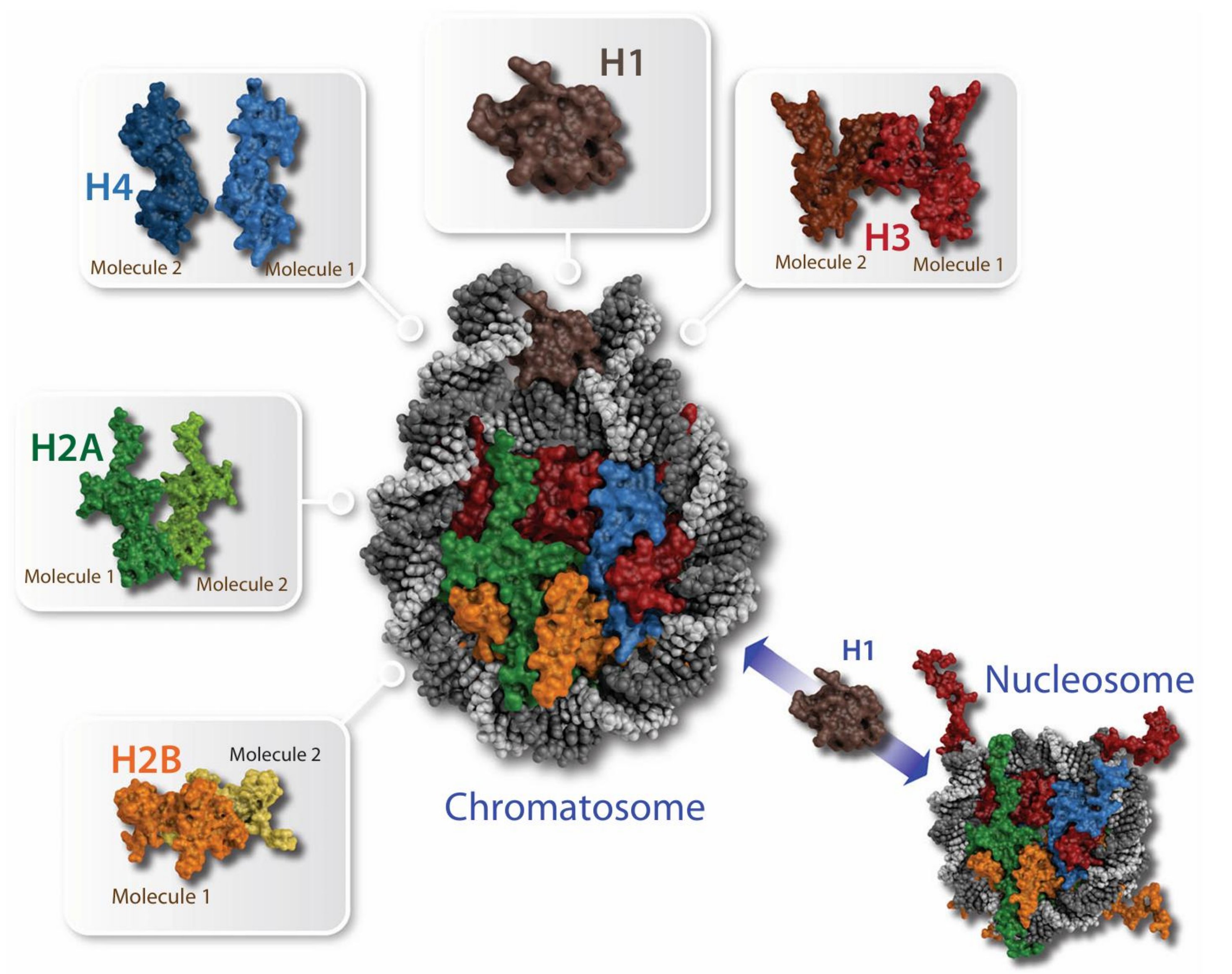
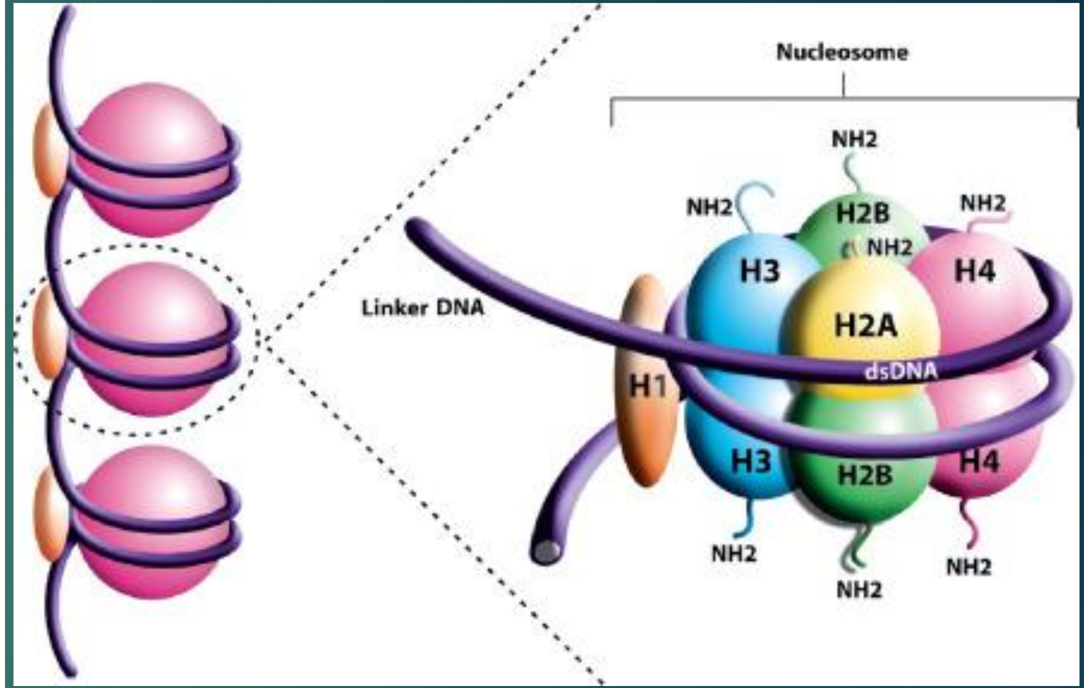
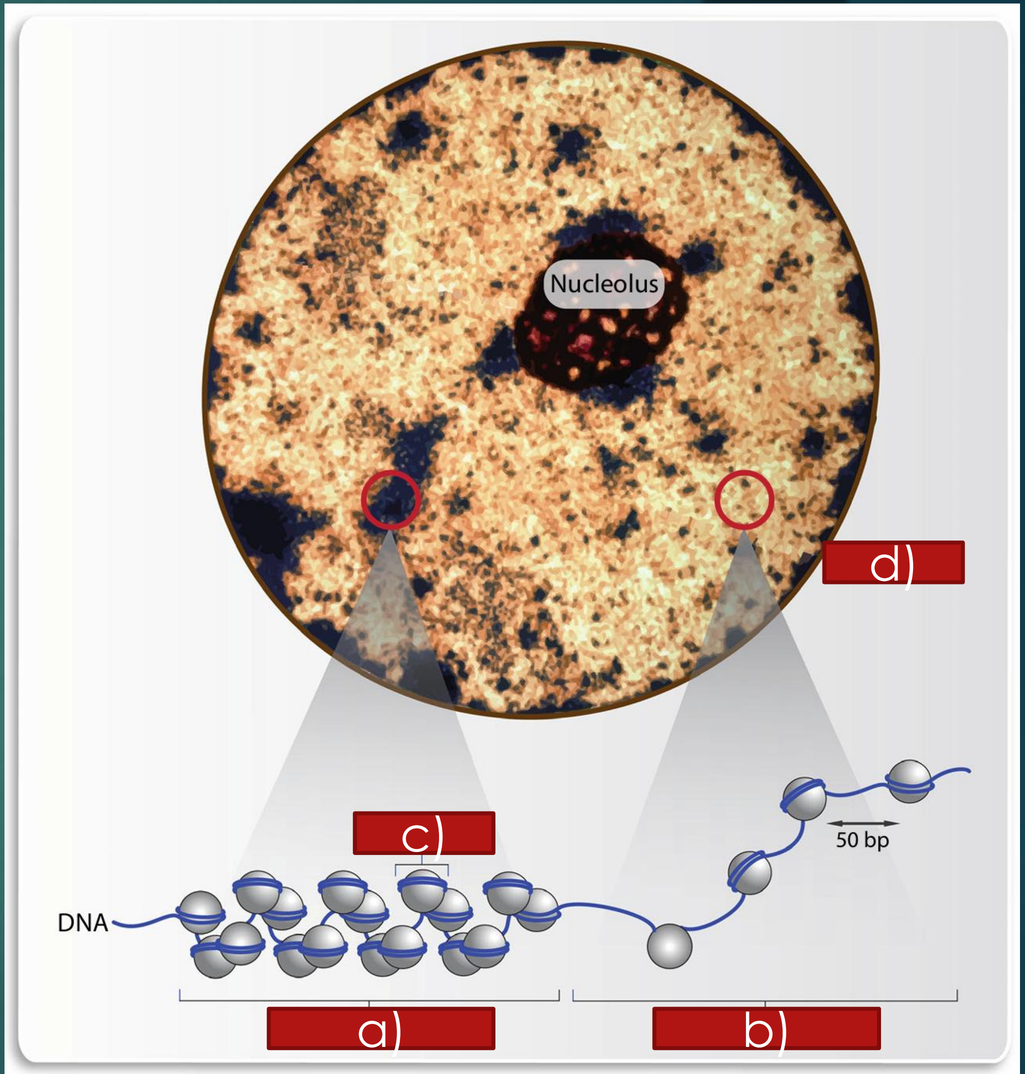




Chromatin in Interphase

- During the non-dividing state (Interphase), the nucleus exhibits two main chromatin states based on condensation:
  - **Euchromatin** (loosely packed, active)
  - **Heterochromatin** (tightly packed, silenced)
- These states are visible via microscopy and are the physical manifestation of epigenetic status.



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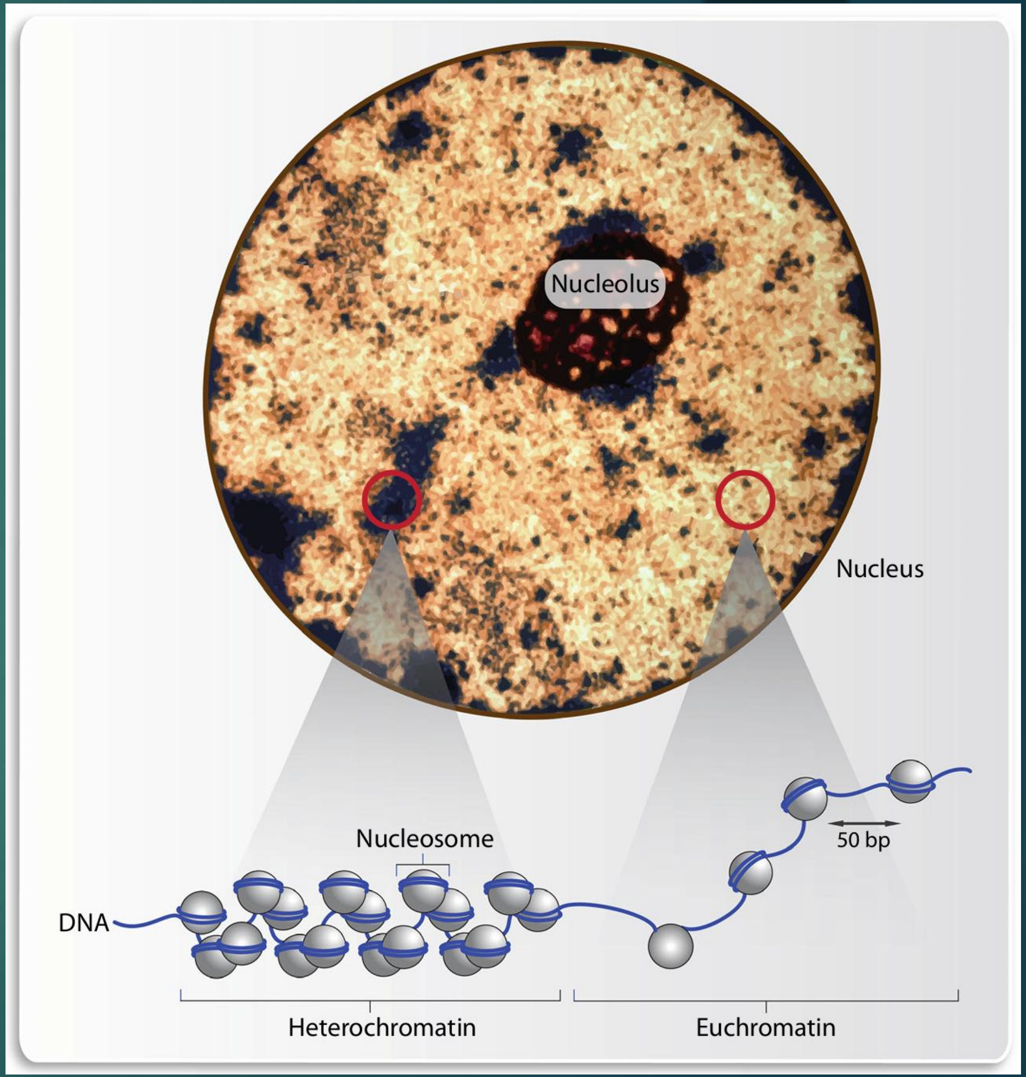
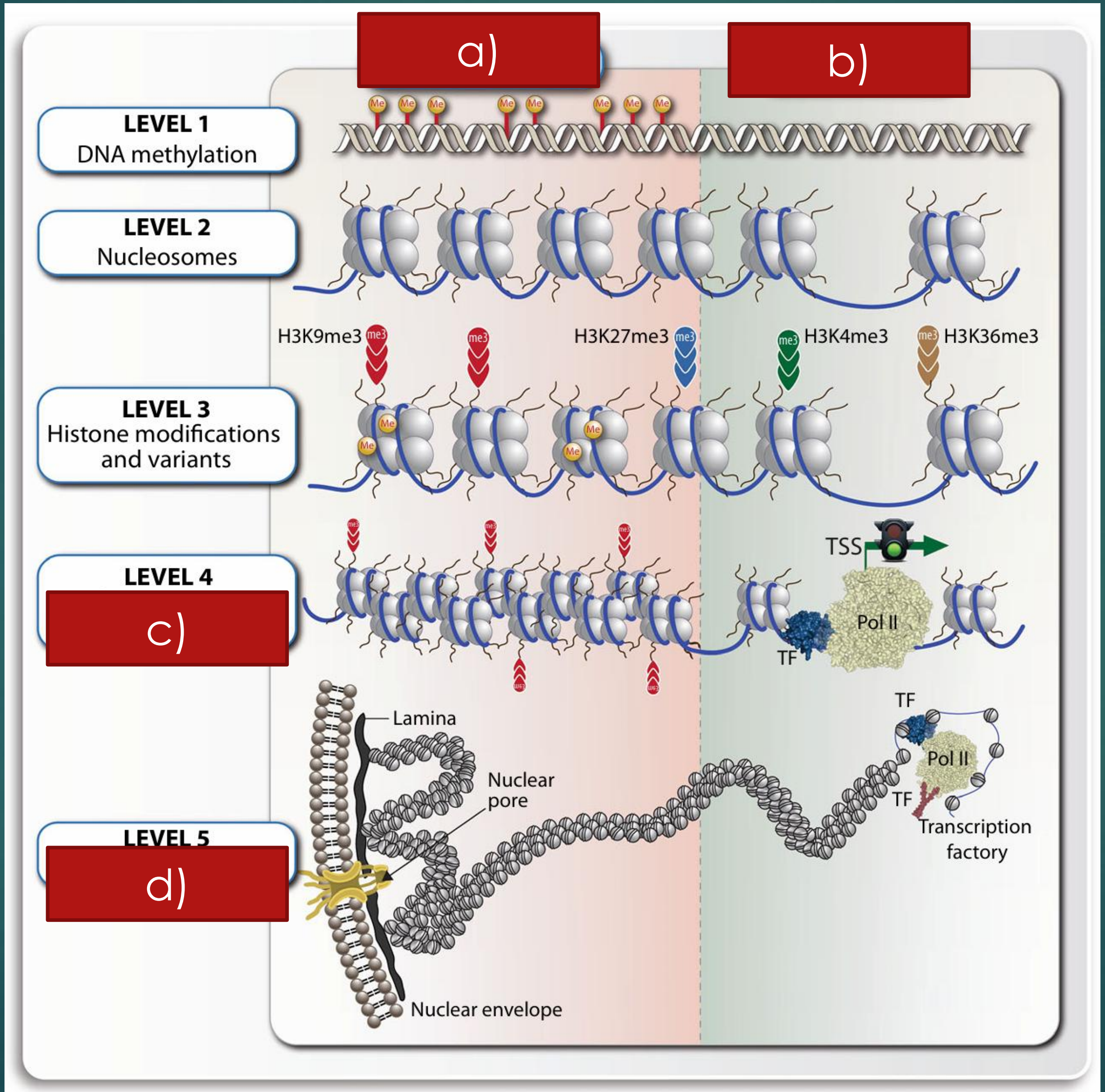
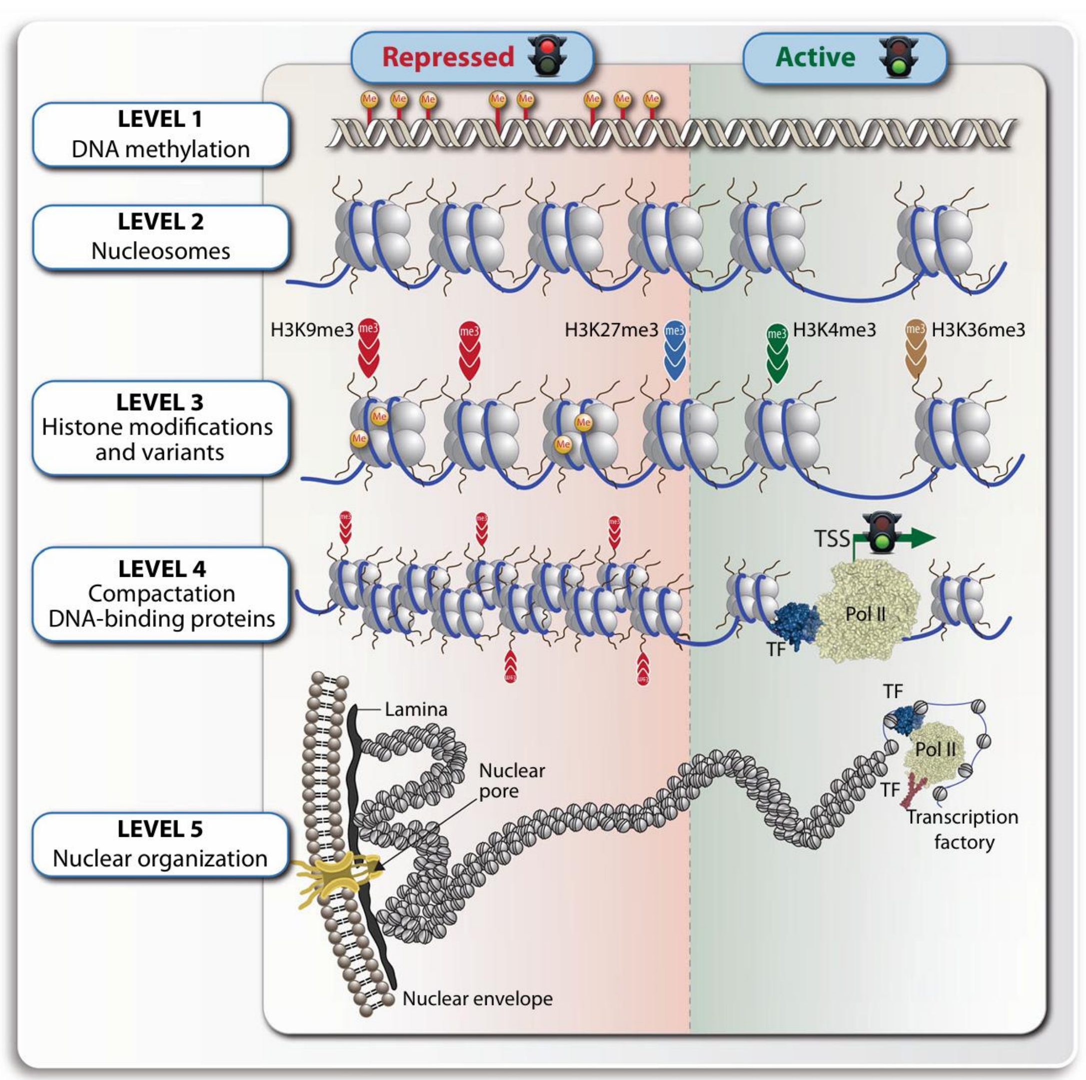


Nuclear Periphery and Silencing



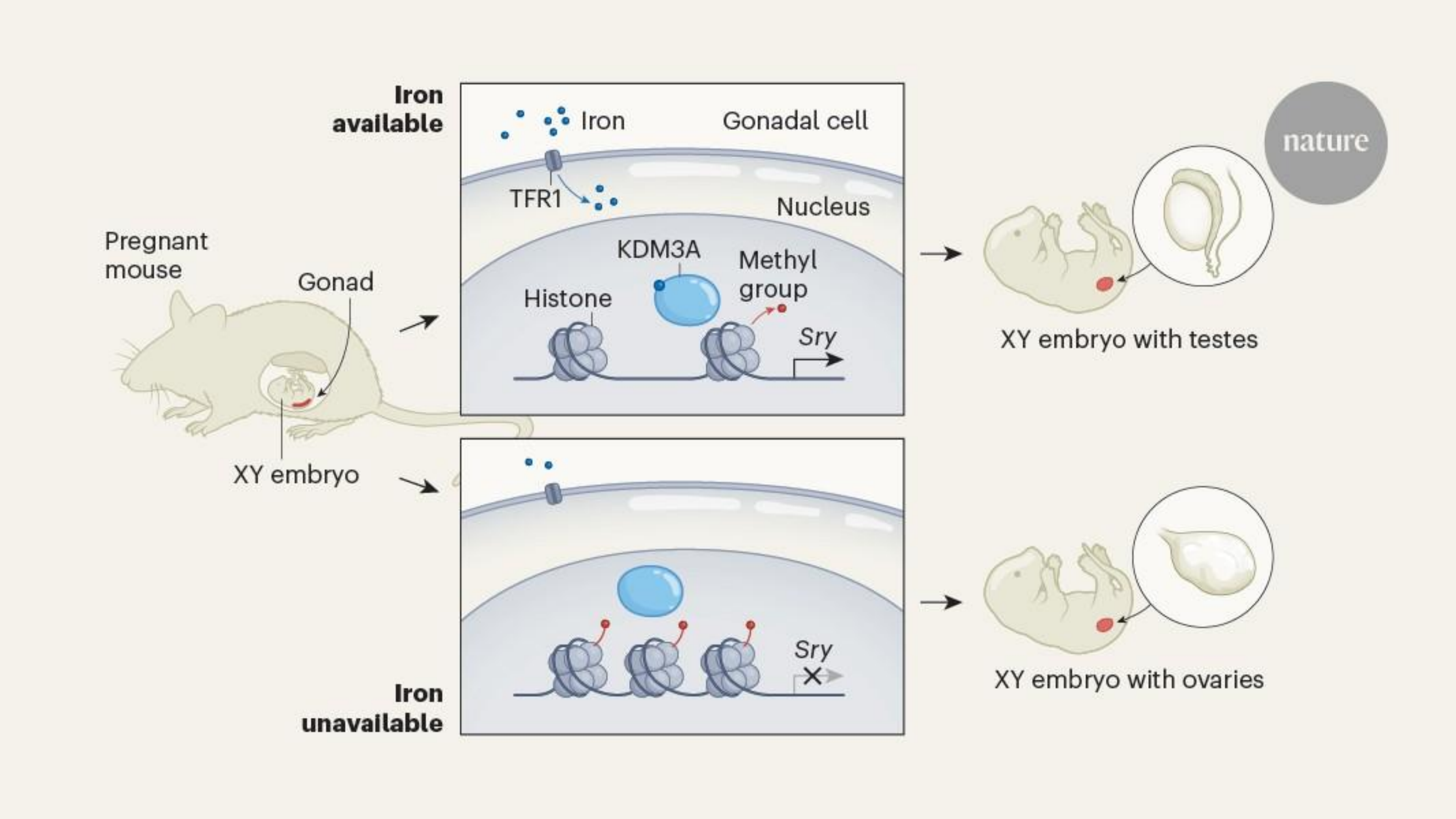
Nuclear Periphery and Silencing

- **Nuclear Lamina**: A protein meshwork beneath the inner nuclear membrane.
- **Lamin-Associated Domains (LADs)**: Large regions of heterochromatin that are physically anchored to the Nuclear Lamina.
- **Consequence**: Tethering to the periphery is strongly associated with stable, long-term gene silencing.



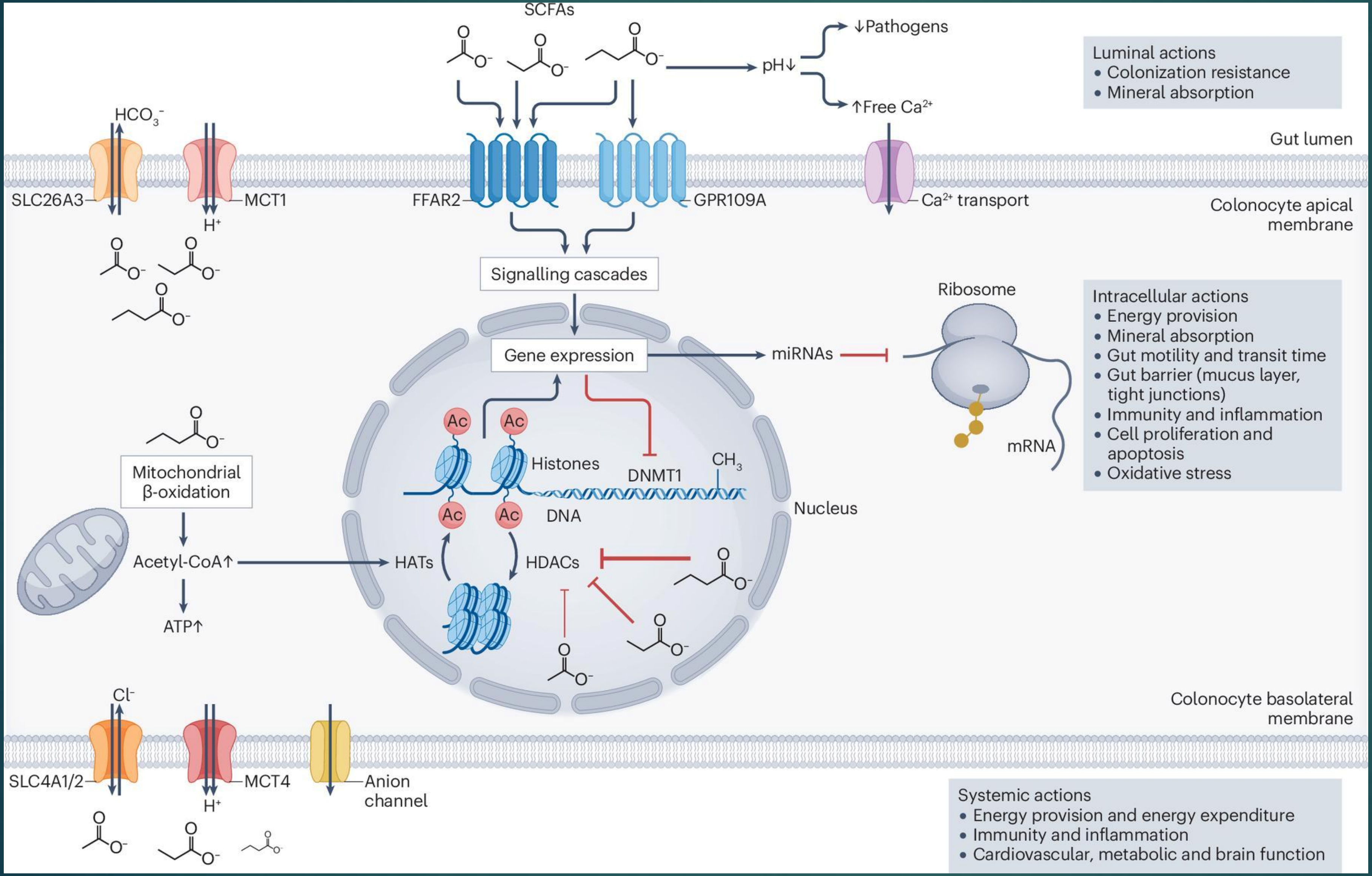
Interpretation

In mice, a lack of maternal iron impairs an iron-dependent enzyme that activates the male sex-determining gene, causing some XY embryos to develop ovaries



KDM3A is a histone lysine demethylase, meaning it removes methyl groups from lysine residues on histones. Specifically, it demethylates H3K9me2 and H3K9me1, making it a crucial enzyme for regulating gene expression and chromatin structure.

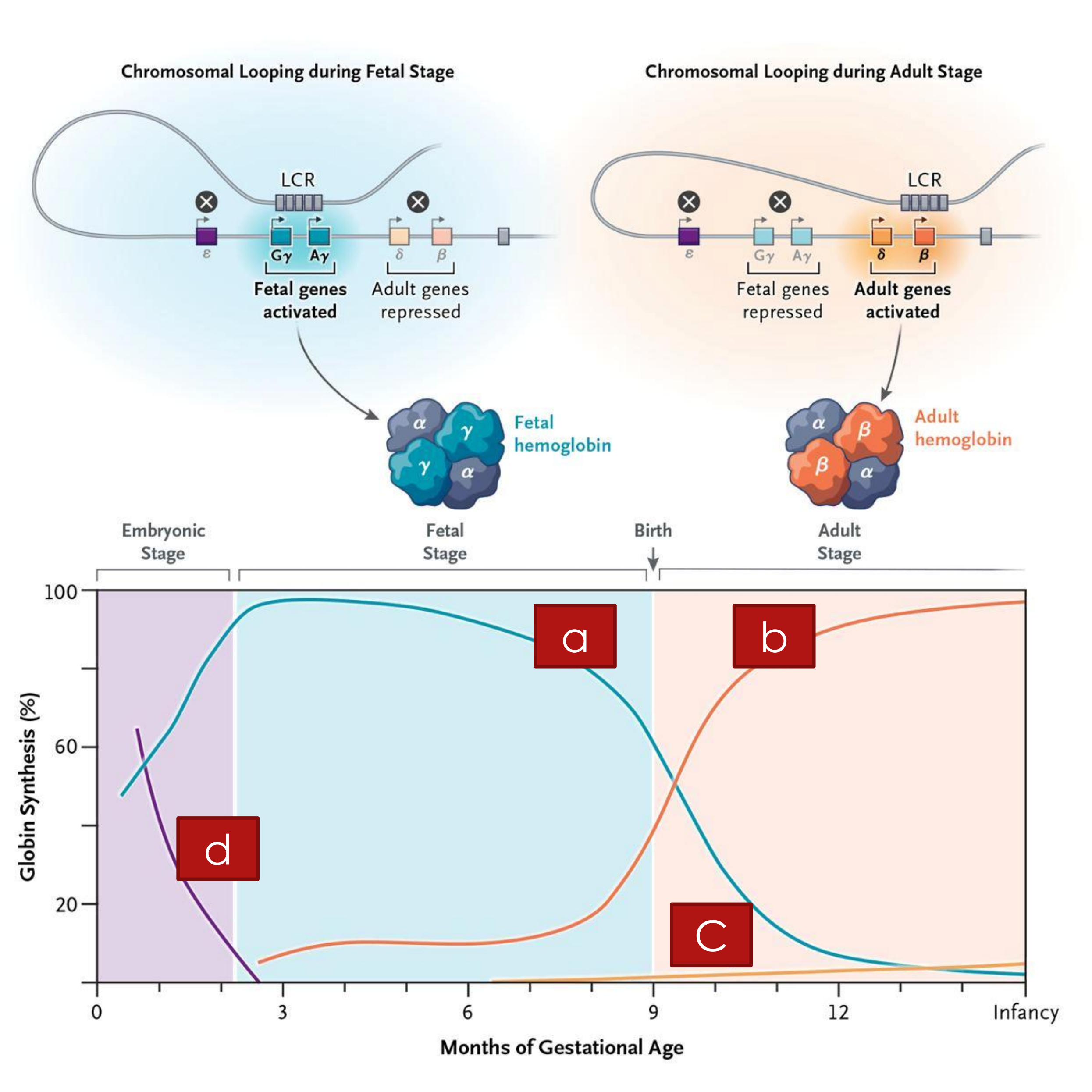
Gut microbiota-derived short-chain fatty acids and their role in human health and disease



Short-chain fatty acids (SCFAs) are a group of organic compounds produced by the fermentation of dietary fibre by the human gut microbiota. They play diverse roles in different physiological processes of the host with implications for human health and disease. This Review provides an overview of the complex microbial metabolism underlying SCFA formation, considering microbial interactions and modulating factors of the gut environment. We explore the multifaceted mechanistic interactions between SCFAs and the host, with a particular focus on the local actions of SCFAs in the gut and their complex interactions with the immune system. We also discuss how these actions influence intestinal and extraintestinal diseases and emerging therapeutic strategies using SCFAs.

https://www.nature.com/articles/s41579-025-01183-w

The Fetal-to-Adult Hemoglobin Switch — Mechanism and Therapy



Locus Control Region (LCR)

DOI: 10.1056/NEJMra2405260

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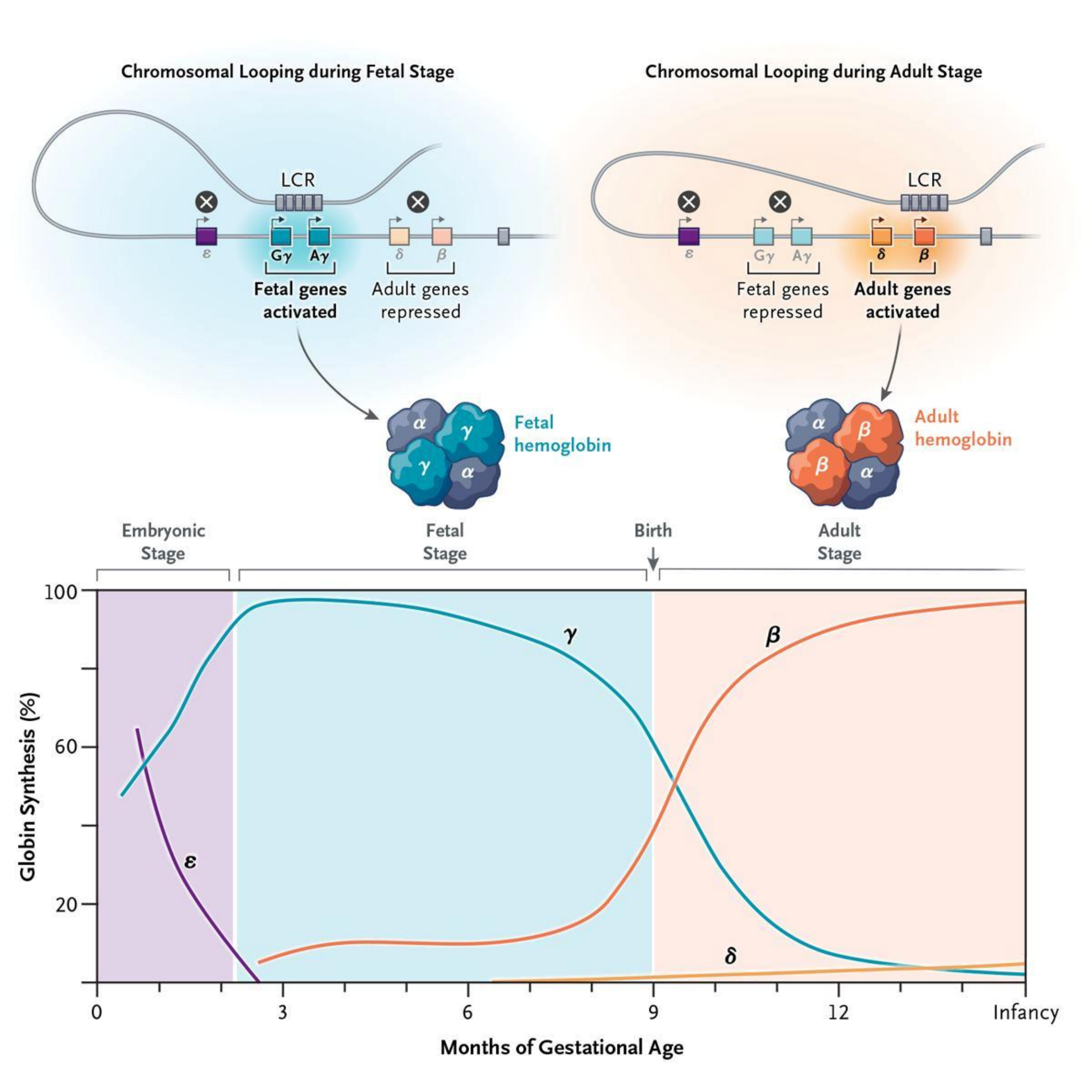

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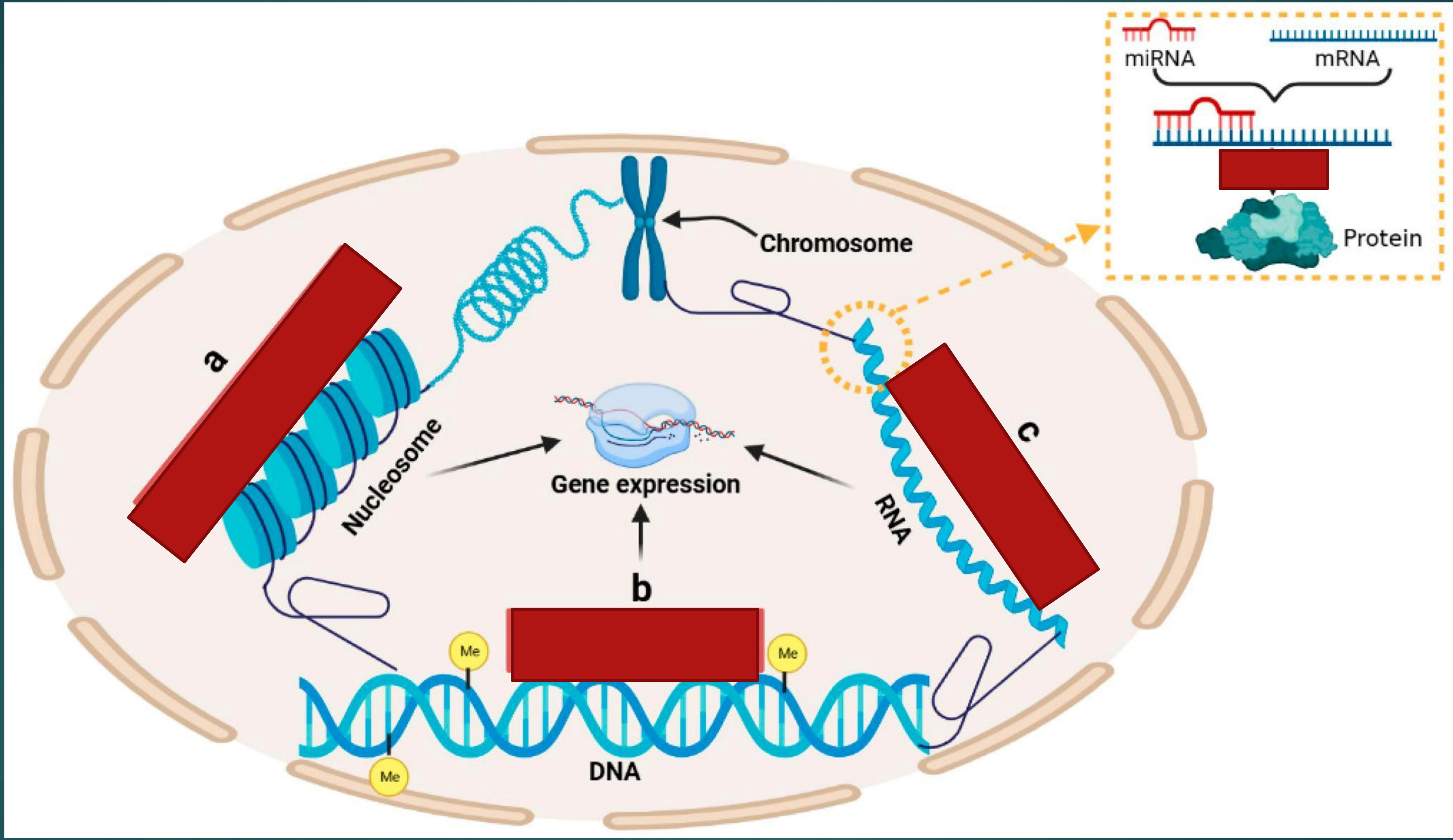
Epigenetic Mechanisms in Gene Regulation



Zhang, J.; Sheng, H.; Hu, C.; Li, F.; Cai, B.; Ma, Y.; Wang, Y.; Ma, Y. Effects of DNA Methylation on Gene Expression and Phenotypic Traits in Cattle: A Review. Int. J. Mol. Sci. 2023, 24, 11882.

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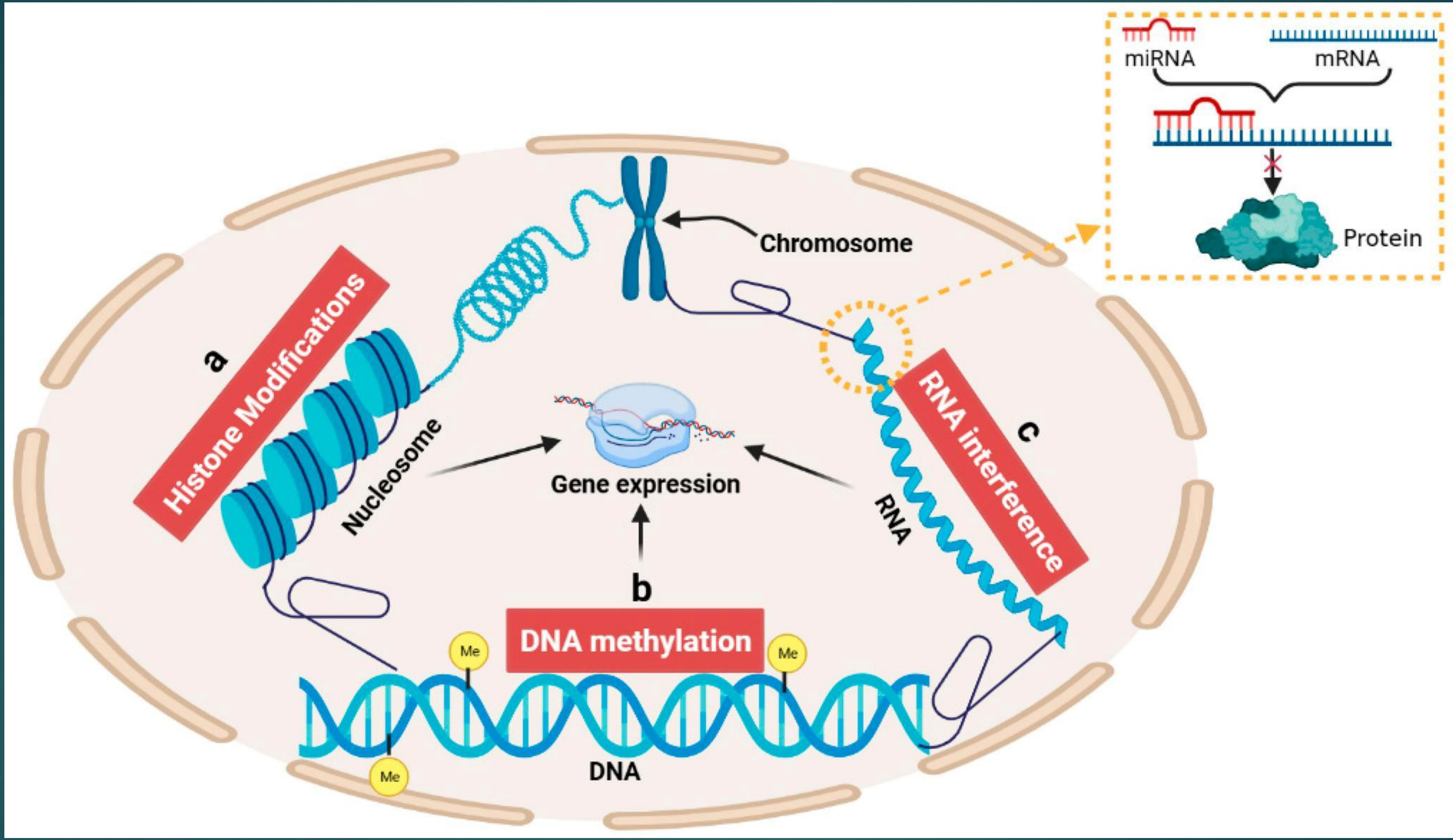

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Thank you for your time!

Phuc-Loi Luu, PhD

Email: luu.p.loi@googlemail.com

Zalo: 0901802182